



Summary of this week

- Congratulations for completing this course!
- You have now learned how the particle filter is derived to approximate the brute-force Bayesian solution and how it solves the “curse of dimensionality” for highly nonlinear state-estimation problems.
- You have learned how to implement a particle filter in Octave, including how to resample the particles if they become degenerate.
- You have seen applications to state estimation and to indoor navigation.
- I hope that you now see that (even though its derivation is challenging) the particle filter itself is a very simple thing—the challenge to implementing one for any specific application is determining the conditional pdfs $f(z_k | x_k^{(i)})$ and $f(x_k^{(i)} | x_{k-1}^{(i)})$ and choosing the importance density $q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)$.



Where to from here?

- If you have studied all four courses in this specialization, you now have a very solid foundation for exploring an enormous variety of Kalman-filter theory and applications of the Kalman filter.
- You have learned the fundamental concepts required to understand alternate derivations and forms of the Kalman filter and various enhancements (especially to nonlinear Kalman filters) that different authors have proposed.
- You have learned how to implement Kalman filters in Octave for example scenarios relating to state-estimation, target-tracking, simultaneous state and parameter estimation, and navigation.
 - You are well prepared to apply Kalman filters to other scenarios relating to these applications, and perhaps others I have not even thought about!
- I wish you great success as you apply the material from this specialization to the problems you seek to solve.